Part A — Armington with a productivity spillover

Point (a)

Set-up. As in Topic 6, location i produces one variety with labour only, $Y_i = A_i L_i$, under perfect competition. The mill price is $p_i = w_i/A_i$ and the price paid in destination n is $p_{ni} = \tau_{ni} w_i/A_i$. The only change is the production spillover

$$A_i = \bar{A}_i \, L_i^{\lambda}, \qquad \lambda \geq 0, \tag{1}$$

so that the delivered price becomes

$$p_{ni} = \frac{\tau_{ni} \, w_i}{\bar{A}_i L_i^{\lambda}}. \tag{2}$$

Price index. With CES preferences over the N varieties,

$$P_n^{1-\sigma} = \sum_{i=1}^{N} p_{ni}^{1-\sigma} = \sum_{i=1}^{N} \phi_{ni} \left(\frac{w_i}{\bar{A}_i L_i^{\lambda}} \right)^{1-\sigma} = \sum_{i=1}^{N} \phi_{ni} \, \bar{A}_i^{\sigma-1} \, L_i^{\lambda(\sigma-1)} \, w_i^{1-\sigma}, \qquad \phi_{ni} = \tau_{ni}^{1-\sigma}. \tag{3}$$

Trade shares. The share of n's spending on goods from i is

$$\pi_{ni} = \frac{p_{ni}^{1-\sigma}}{P_n^{1-\sigma}} = \frac{\phi_{ni} \left(w_i/\bar{A}_i L_i^{\lambda} \right)^{1-\sigma}}{\sum_k \phi_{nk} \left(w_k/\bar{A}_k L_k^{\lambda} \right)^{1-\sigma}}. \tag{4}$$

Trade balance. Goods-market clearing: the labour income of i equals the spending of all destinations on i's variety,

$$w_i L_i = \sum_{n=1}^{N} \pi_{ni} \, E_n, \tag{5}$$

where E_n is n's spending on goods. With the convention of Topic 6 (spending on goods equal to labour income, $E_n = w_n L_n$),

$$w_i L_i = \sum_{n=1}^{N} \frac{\phi_{ni} \left(w_i/\bar{A}_i L_i^{\lambda} \right)^{1-\sigma}}{\sum_k \phi_{nk} \left(w_k/\bar{A}_k L_k^{\lambda} \right)^{1-\sigma}} \, w_n L_n. \tag{6}$$

Population now appears on both sides: on the left as the size of the labour force, on the right through productivity, with exponent $\lambda(\sigma - 1)$.

$\bar{A}_i$ and λ enter only through A_i. In (3)–(6) the fundamental $\bar{A}_i$ and the elasticity λ never appear separately: they appear only in the product $\bar{A}_i L_i^{\lambda}$, that is in A_i. The other equilibrium conditions of Topic 6 — housing-market clearing, real utility and the Fréchet migration shares — do not contain productivity at all, except through P_n and w_n. Therefore the whole equilibrium (w, L, P, π, r) depends on $(\bar{A}, \lambda)$ only through the composite A: two parametrizations with the same $A_i = \bar{A}_i L_i^{\lambda}$ in every location are the same economy.

Inversion. Given data (L_i, w_i, H_i), the trade costs and σ, the inversion of Topic 6 treats the trade-balance condition (6) as N equations in the N unknowns w_i/A_i, i.e. in A_i. Trade shares depend only on relative values of w_i/A_i, so A is determined up to a common scale, which a

normalization fixes; up to that normalization the solution is unique. The inversion uses L and w as data and never needs λ: it recovers the composite A_i *exactly* (`inv.Aeff` in the code).

To go from A_i to the fundamental we need λ:

$$\bar{A}_i = A_i \, L_i^{-\lambda}. \tag{7}$$

So $\bar{A}_i$ is recovered only *for a given* λ (`inv.A`).

Every λ fits the data equally well. Take any $\lambda' \geq 0$ and set $\bar{A}_i(\lambda') = A_i L_i^{-\lambda'}$. Then $\bar{A}_i(\lambda') L_i^{\lambda'} = A_i$ in every location, so by the previous paragraph the price indices, trade shares and trade balance are identical to those of the true economy: wages and populations are reproduced exactly, whatever λ' is. One cross-section gives N equations that pin down N composites; splitting them into N fundamentals and one λ needs information that the cross-section does not contain. As in Part C of Homework 1, the data identify A, not $(\bar{A}, \lambda)$.

Why it matters. λ does not change the fit, but it changes the counterfactuals: after a shock to $\bar{A}_j$, workers move, L_j changes, and with $\lambda > 0$ productivity $A_j = \bar{A}_j L_j^{\lambda}$ moves with them. The same data inverted with different λ therefore give different answers to the same policy question (point (c)).